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A Novel Approach for Character Segmentation of Offline Handwritten Marathi Documents written in MODI Script

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Abstract

One of the most challenging module in any handwritten character recognition system is separating individual characters from the document. Better character recognition rate is highly influenced by how much accuracy, character segmentation phase, has achieved. As the input data set is contains documents written in MODI script, and since MODI script characters are written in cursive manner, obtaining individual characters from the text becomes extremely complicated. Moreover, there is no separation between two adjacent words. Although now-a-days, Marathi language uses Devnagari script, several years back MODI Script was used for writing documents in Marathi language. .This work presented in this paper tackles segmentation of individual characters from ancient handwritten MODI Script documents. Vertical Projection Profile (VPP) method can segment characters from a given line accurately only when two adjacent characters are separated by a zero-pixel column. Since characters of one line in this script are written without lifting hand, output of VPP does not suffice. This paper presents a novel approach for isolating individual characters from the line obtained using authors' previous research work which uses dual thresholding criterion to minimize the character segmentation error. The techniques used in this paper are straightforward and reasonably economical in terms of execution time efficiency. The results obtained are quite satisfactory.

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Keywords: MODI Script; binarization; VPP; dual thresholding criterion; character segmentation

1. Introduction

MODI script character segmentation is one of the most challenging phase, as there is hardly any separation among the adjacent characters written in these documents. Entire line of these documents is written without lifting one's hand. This paper deals with isolating characters from the inputted line.
The organization of this research paper is given below: Section 2 discusses the state-of-the-art, particularly focusing on character segmentation work. Authors' previous research work is described in Section 3. Section 4 elaborates the actual work carried out for this paper. Then, in section 5, experimental results are discussed. Conclusion and future directions are discussed in section 6. Lastly, the paper ends with acknowledgments and references in section 7 and 8 in that order.

2. State-of-the-Art

This section highlights the work done by various researchers on character segmentation on different scripts.

[2] presents an approach to identify width of a stroke of an individual pixel by applying morphological thinning process on the words within the documents, thereby, causing over-segmentation,
Gurumukhi script character segmentation is carried out in [3] using water-reservoir method. It works even though the adjacent characters are touching each other.
Word segmentation is performed in [4] using an unsupervised learning approach. The script used is cursive in nature. Touching locations are identified using self-organizing maps.
A rigorous survey is carried out in [5] on different segmentation techniques, particularly focusing on character segmentation. Mostly, characters from Indian scripts are chosen. The cursive characters are also considered.
Conscience Online Learning approach is used in [6] for the purpose of segmenting individual characters. The results obtained are quite satisfying because of the nonlinear nature of the spectral clustering method used. Two different methods used are kernel clustering and normalized cut clustering.
[7] performed character segmentation work on handwritten Chinese script. To find out probable candidate segmentation points, Viterbi algorithm is used. Due to its probabilistic nature, nonlinearity is achieved.
Handwritten Hindi overlapping characters are used in [8]. Various approaches are tried for their efficient segmentation.
[9] deals with segmentation and recognition of two adjacent digits using fuzzy logic approach. It identifies connected components from the image and represents them in fuzzy terminology. The segmentation yields good results due to the uncertainty captured by fuzzy sets.
Touching and/or overlapping Chinese characters are segmented using [10] by using a Genetic algorithm approach. Possible segmentation points are calculated appropriately by the heuristic approach presented in the paper.
MODI script character segmentation work is carried out in [12] using horizontal zoning approach. Over segmentation and/or under segmentation is observed due to highly complex nature of the script. Moreover, multiple phases are used to obtain the desired results. Whereas, the work presented in this paper performs character segmentation in a single step technique.

3. Previous Research Work

The summary of the previous research work is described below:
Step 1: MODI script document images are scanned/acquired with a spatial resolution of 300 dpi (Figure 1).
Step 2: Each input image is converted from RGB to gray-scale using rgb2gray function (Figure 2).
Step 3: Gray-scale images obtained from step 2 are resized to a square matrix of size 1024x1024 pixels.

Step 4: Show-through effect in these highly degraded MODI script document images is largely alleviated by preserving most of the text image details by analyzing histogram peaks and valleys obtained using [1], thereby, selecting an appropriate global threshold value for image binarization.
Step 5: The sample images are binarized using the threshold obtained in the previous step.
Step 6: Line segmentation work is carried out using [11].
Step 7: For each line extracted using step 6 is taken as an input for the current research work explained in section 4.

4. The Proposed Method

4.1 Problem Statement

Let L = L(x) be a segmented line image resulted from the previous research work, where x = [i, j]$^T \in \boldsymbol{R^2}$ with 'r' rows and 'c' columns. The goal is to obtain individual characters accurately segmented from the line with minimal interference of letter modifiers.

4.2 Mathematical Formulation:

A new augmented matrix L_D of size $(r+1)x\,(c+1)\ where\ 1^{st}\ row, (r+1)^{th}\ row, 1^{st}\ column, and\ (c+1)^{th}\ column\ contains\ all\ 0"s$, which is shown as below:

$$L_D = \begin{bmatrix} 0 & 0 & 0 \\ 0 & L & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Assume w to be a matrix of size 3 x 3 containing all 1's as shown below:

$$w = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

For every point (i,j) in x of $L(x)$,

$$,N(x) = \begin{bmatrix} i-1,j-1 & i-1,j & i-1,j+1 \\ i,j-1 & i,j & i,j+1 \\ i+1,j-1 & i+1,j & i+1,j+1 \end{bmatrix}$$

represents 3x3 neighborhood of point (i,j).
It is known that 2-D translation by a point (i_0, j_0) of $\boldsymbol{R^2}$ is given by $T(i,j) = (i+i_0, j+j_0)$. Since translation is not linear, it cannot be represented by 2x2 matrix transformation. So, to write it as a matrix transformation, Affine co-ordinates are used. For any point (i,j) in $\boldsymbol{R^2}$, its Affine co-ordinates are given by (i,j,1). Using translation, T is represented as 3x3 matrix transformation as shown below:

$$\begin{bmatrix} i+i_0 \\ j+j_0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & i_0 \\ 0 & 1 & j_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} i \\ j \\ 1 \end{bmatrix}$$

Suppose B be the matrix of the same size as L_D as shown below:

$$B = \begin{bmatrix} u & 0 \\ 0 & 0 \end{bmatrix}$$

where $u = [1 \quad 1 \quad 1]$.

Now, $B\ L_D B^t T_{i0,j0}$ gives sum of products in 3x3 neighborhood of every point in L_D. Whenever this sum of products is equal to $L_D(p)$, then p is considered to have all 8-neighbors having value 0.

Pixel intensity of each column j, where $\boldsymbol{1 \le j \le c}$ is formulated as a linear function of L, which is defined as

follows:

$$V(j) = \sum_{i=1}^{r} \boldsymbol{L(x)}$$ where 'i' runs over row indices.

Now, the Difference Operator ∇ of function V is defined as:

$$\nabla V(j) = V(j+1) - V(j) \text{ for column indices } \mathbf{1} \leq \boldsymbol{j} < \boldsymbol{c}.$$

Let $\mathbb{1}_c$ be a c x 1 column matrix of all 1's as shown below:

$$[1 \quad \ldots \quad 1]^t$$

For any matrix A of size c x m, the product of $\mathbb{1}_c^t A$ gives a row matrix of size 1x m which is the range of the function V. Also the difference operator ∇ of function V can be represented in a matrix form as follows: $\nabla V(j) = \nabla \mathbb{1}_c^t A e_j = \mathbb{1}_c^t A \nabla e_j = = \mathbb{1}_c^t A(e_{j+1} - e_j)$ where e_0 is a column matrix of all 0's and e_j is standard unit vector with j^{th} entry is equal to 1 and all other entries are 0. Suppose t be the threshold value representing average width of possible segmentation candidate and let s is a threshold value representing average variation in consecutive columns in the document.

According to number theory, any two integers a and b are said to be congruent to each other modulo t if t is a divisor of a-b.

Now, let G be a function from $\{1,2,\ldots,c\}$ to $a\ set\ of\ positive\ integers.$

$$G(k) = 1\ if\ V(k) > 0\ and\ V(1) = V(2) = \ldots = V(k-1) = 0.$$

Now $G(q)$ for $2 \leq q \leq c$ is defined recursively as follows:

$$G(q+j) = G(q) + 1\ if\ \nabla V(q + j - 1) > s\ and\ t \leq j$$
$$= G(q), \text{otherwise}$$

It is clear that G is a one-one and onto function from set of indices of columns which separates each possible segmentation candidate to $a\ subset\ of\ \{1,2,\ldots,c\},\ say\ D$. Therefore G^{-1} from D to set of all possible indices of columns which separates each possible segmentation candidate.

4.3 Proposed Algorithm

The outline of the algorithm is as follows:

Step 1: Initially number of rows and number of columns in the given input line image, say line, is obtained.

Step 2: Then, sum all nonzero values from each column is calculated and stored it in a column vector, say p.

Step 3: Assuming that none of the input line images will contain more than 50 characters, an array of size 50 containing all zeros, say y, is defined.

Step 4: Index of first nonzero entry from vector p is copied as the first element of array y. This index considered as the start of the first character in the given line.

Step 5: End of the character is identified by scanning subsequent values from the vector p and by using the following two conditions:

i) if current cell value from vector p is 0 or,
ii) analyzing the absolute difference between current and previous value from p to be greater than threshold value 1 and the difference between current index value in p and the last index value stored in y exceeds average character width (threshold value 2) of the character.

Step 6: Step 5 is repeated until end of the last character is obtained.

Step 7: All the character indices obtained using previous steps and stored in array y are extracted and stored in the output folder.

5. Experimental Results and Discussion

The code is tested on 66 different historical MODI script documents containing more than 790 lines with an average of 20 characters per line.
One sample RGB image of a MODI script document is shown in figure 1. The bleed-through effect is evidently perceptible. Figure 2 depicts gray-scale image. Figure 3 contains binarization result obtained using [1]. Figure 4 shows line segmentation result obtained using [11]. The line image inputted to the algorithm presented in this research paper is shown in figure 5. The output of the current research work is shown in Figure 6.

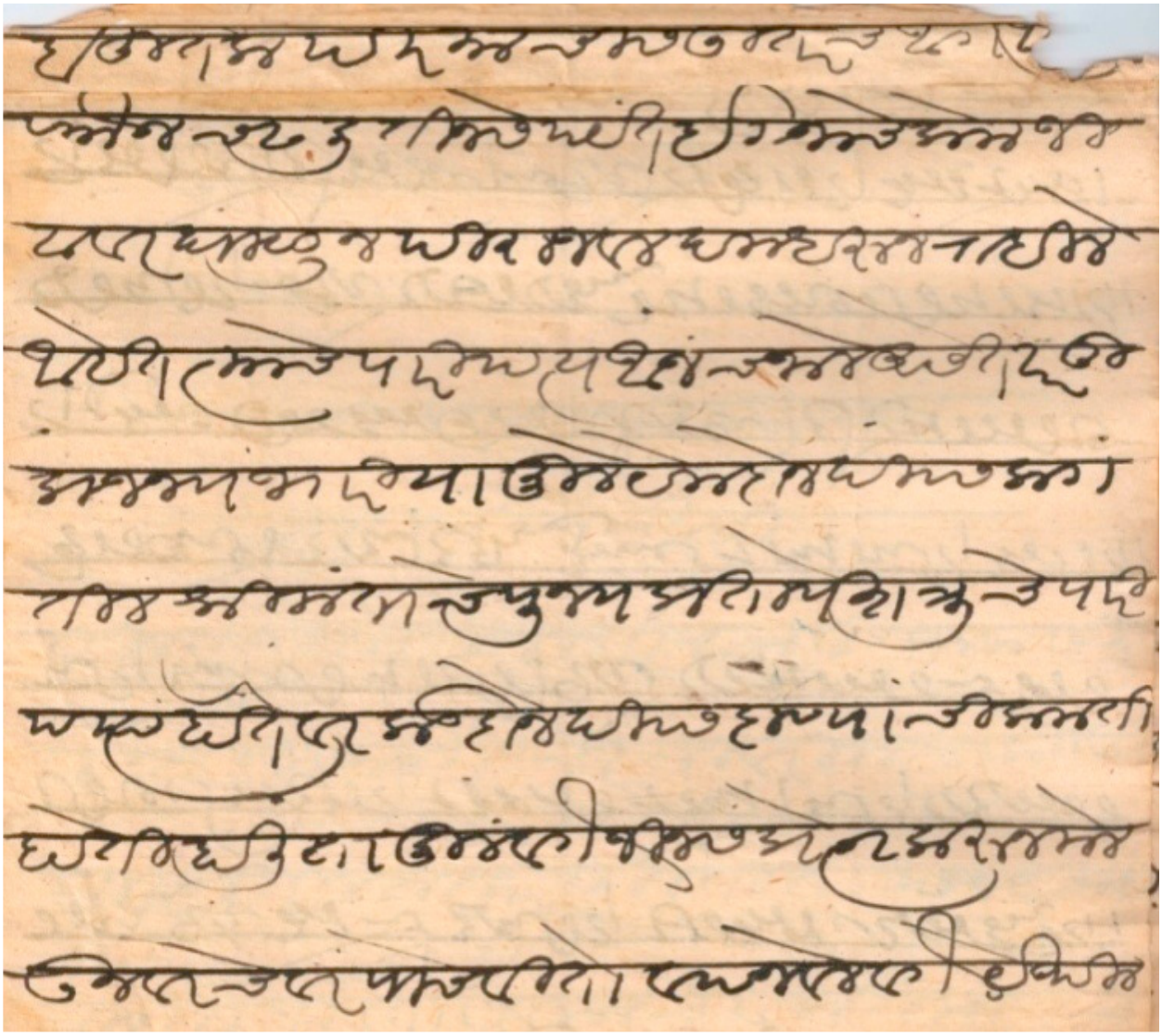


Figure 1: One sample Original RGB Image

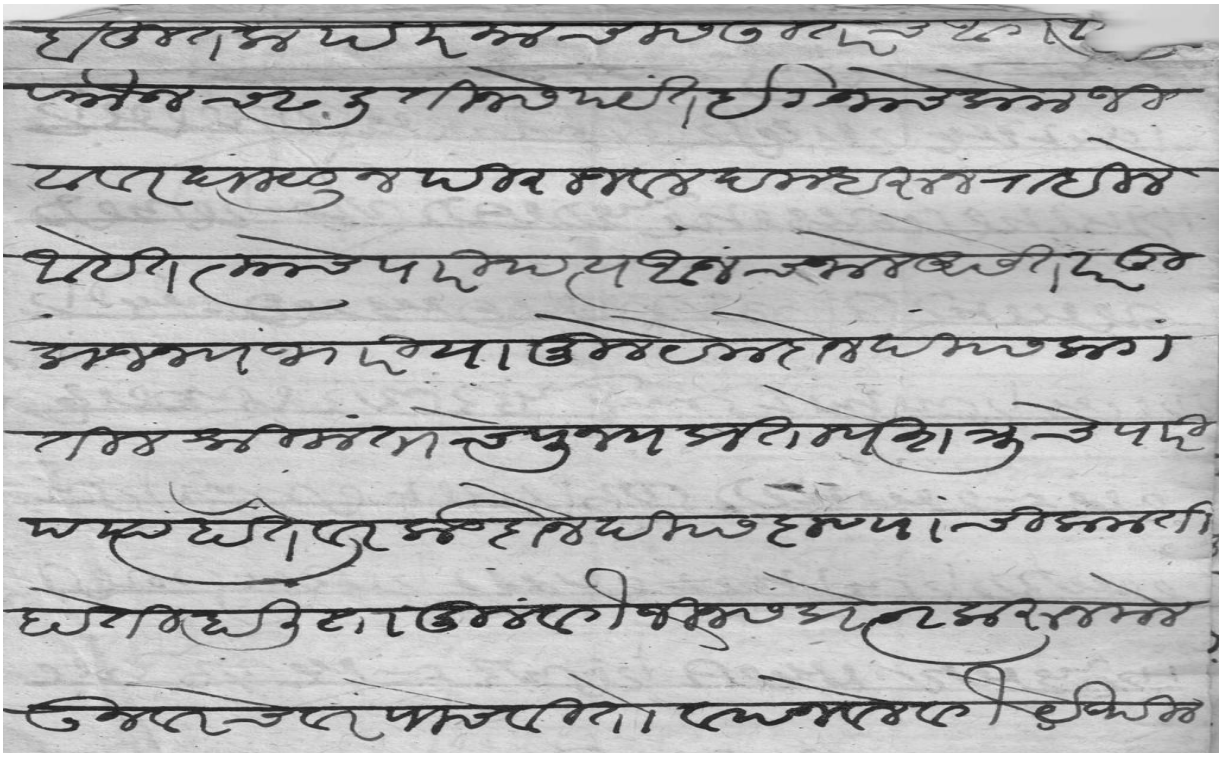


Figure 2: Gray-Scale Image



Figure 3: Binarized and padded image using [1]

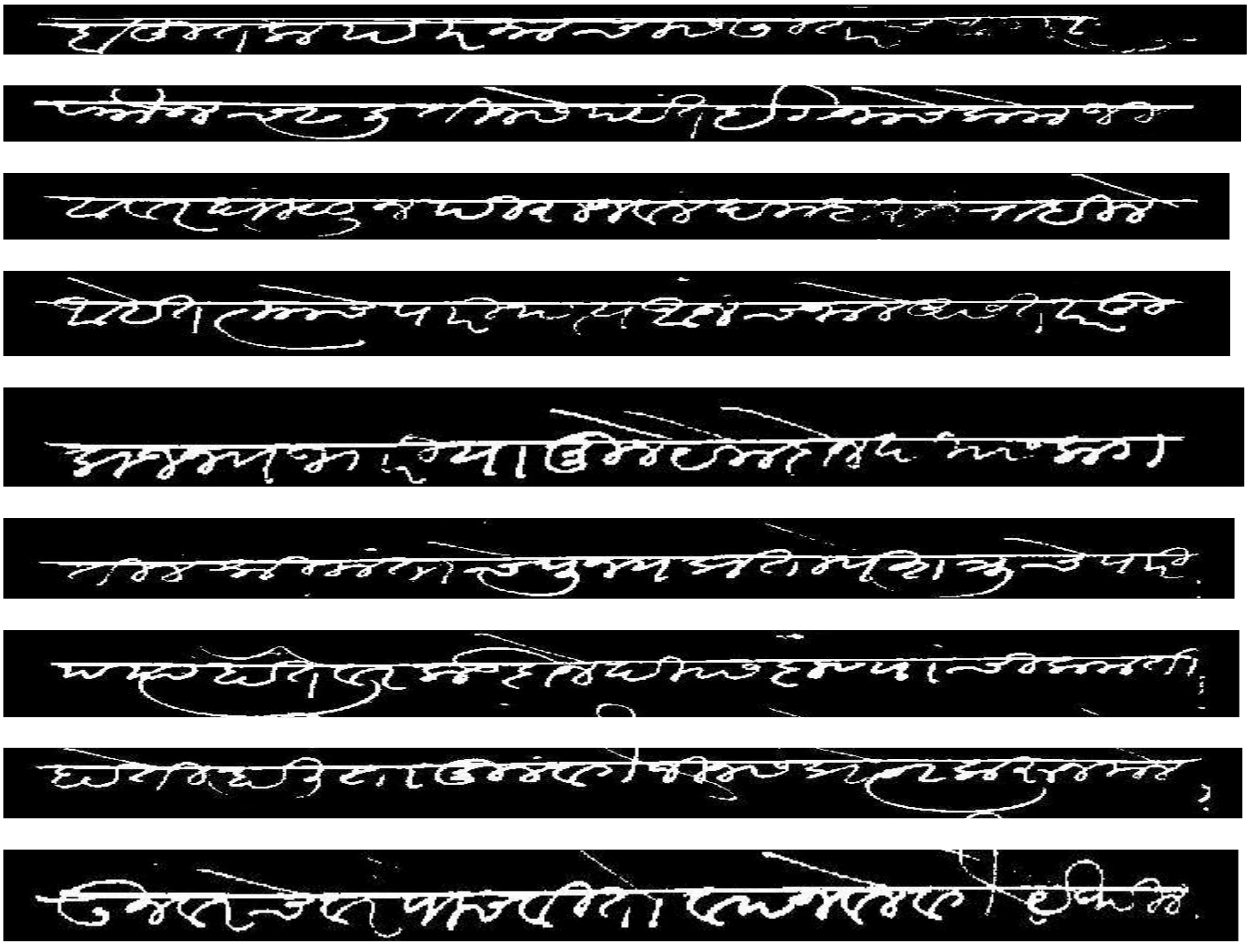


Figure 4: Lines Segmentation Result obtained using [11]

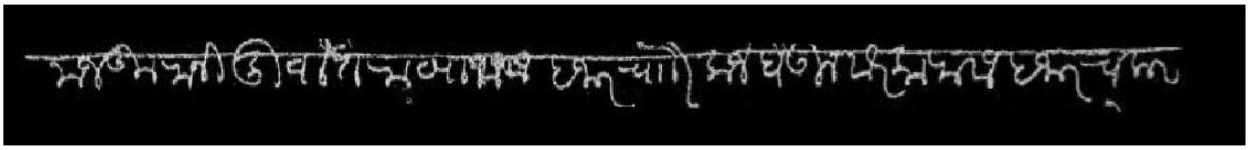


Figure 5: A sample input line image





Figure 6: character Segmentation Result obtained using the proposed algorithm

6. Conclusion and Future Prospects

The algorithm was tested on more than 15000 characters written by different writers with varied writing styles, out of which almost 10000 characters were segmented accurately, thereby, giving an accuracy of approximately 67%.
The results obtained using the algorithm presented in this research paper yields good results provided the width of the character is not too small or not too large.
Future work involves improving upon the current success rate to correctly segment the characters in the given line, and then perform character recognition using deep learning approach.

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